Racial Segregation and Black-White Longevity Disparities*

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Abstract

Racial segregation is widely understood to be a mechanism linking racial stratification to zero-sum Black-White outcomes. However, historical and demographic evidence show that segregation also shapes broader, macro-level dynamics, such as income sorting, the availability of public goods, and policy outcomes that undermine the well-being of all residents. This paper tests the argument that segregation leads to White advantage and, simultaneously, Black disadvantage. To do so, I link 1905-1920 birth cohorts in the 1940 Census to Social Security death records and use an instrumental-variable design leveraging variation in government fragmentation to estimate the effect of segregation on longevity. Estimates reveal a 10-point increase in segregation reduced Black and White longevity by 0.44 and 0.29 years, respectively, with pronounced harms for less-educated Whites. These findings highlight how mechanisms of racial stratification intersect with population processes and social class to shape life chances across racial groups, thus countering zero-sum arguments.

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1 Introduction

The persistent segregation of Black Americans (Elbers 2021) is both a mechanism of structural racism (Brown and Homan 2024; Massey and Denton 2003) and a fundamental dimension of Black-White inequality (Blokland 2008). A central assumption in scholarship and public opinion is that Whites benefit from racial segregation and will lose advantages afforded to them if they were to residentially integrate with Black Americans. Zero-sum views about Black-White relations are pervasive and form the basis of Whites' (Warren and Valentino 2025; Wetts and Willer 2018) public opinions about redistributive policy (Gilens 1996) and racial integration (Bobo and Zubrinsky 1996). Accordingly, Whites have continually resisted residential and educational integration due to a belief that segregation produces White advantage (Crowder and South 2008; Frey 1979; Johnson 2019; Logan, Zhang, and Oakley 2017; McGhee 2021; Quillian 2014; Rich n.d.).

Despite the ostensible appeal of zero-sum frameworks, historical evidence and demographic processes question the notion that racial stratification advantages all Whites. First, DuBoisian historical analysis describing the changing forms of racial inequality in the Reconstruction era (1935) argues that racial stratification benefits Whites primarily through symbolic mechanisms, such as perceptions of high status, that serve to offset material disadvantages, particularly for lower-class Whites. Second, sociological theory on segregation centers on the role of neighborhood-level differences (Quillian 2014), which are thought to produce zero-sum outcomes; However these perspectives overlook the common effects of macro-level processes that affect all residents. For example, demographic evidence shows that segregation also shapes income segregation (Reardon and Bischoff 2011), public spending on infrastructure and public goods (Chyn, Haggag, and Stuart 2023), and the political efficacy of policies supporting low and middle-income families (Ananat and Washington 2009) that could offset neighborhood-level differences.

In this paper, I bridge historical evidence and theory on racial stratification and empirically test the argument that segregation produces zero-sum outcomes between White and Black Americans. To do so, I examine the effects of racial segregation on the longevity of Black and White Americans using data from early 20th century birth cohorts (1905-1920) in the 1940 Full Count Census linked to Social Security Numident death records (N = 2.1 million). A fundamental challenge in determining the impacts of segregation is the endogeneity of place-level confounders like public policies and infrastructure that impact spatial sorting across places and ultimately affect health outcomes (Montez et al. 2020). To address this, I utilize a validated instrumental variable approach that leverages variation in local government fragmentation, as measured by the number of local governments, within U.S. counties in 1967 (Cutler and Glaeser 1997).[1] The intuition of this approach is that counties with greater government fragmentation experience greater sorting between neighborhoods that increases segregation but is unrelated to individual outcomes.

Results reveal that a 10-percentage point increase in segregation decreased Black longevity by 0.44 years and White longevity by 0.29 years. For Whites, the effects of segregation are particularly harmful for those with lower education; but even for the highest educated White Americans, longevity is still reduced by 0.19 years. These findings highlight how processes of racial stratification affect Whites by virtue of their class position and contribute nuance to straightforward zero-sum conceptions of Black-White inequality. These results contribute to sociological theory by highlighting how mechanisms of racial stratification intersect with structural forces, population processes, and social class position to shape the shared experiences of individuals across racial lines. Taken together, evidence suggests that policies promoting racial integration could improve both Black and White health outcomes.

[1]It is noted in (Cutler and Glaeser 1997) and Light and Thomas (2019) that the spatial arrangement of governments is largely constant over time and fixed well before the contemporary period. This gives additional confidence that contemporary segregation patterns do not shape government location and the degree of government fragmentation within places is not determined by other contemporaneous political or economic forces that may shape individual outcomes. This point is discussed further in the Methods section.

2 Background

2.1 Historical Processes of Racial Stratification and Zero-Sum Views Of Racial Inequality

Although mechanisms of racial stratification such as political disenfranchisement and legal subjugation were targeted at Black Americans, historical record shows that Slavery and Jim Crow laws had significant scope to harm poor Whites as well. As early as the 1850s, observers of chattel slavery in the Antebellum South recognized that processes of systemic racial inequality had profound impacts that were not confined exclusively to Black Americans. In The Impending Crisis in the South: How to Meet, Hinton Rowan Helper outlined an argument for the abolition of slavery, in part because of its economic inefficiencies and disenfranchisement of poor, non-slaveholding Whites. Helper demonstrated that free, Northern states had surpassed the South in trade, production, and economic diversity (Helper 1857:26). Though slavery explicitly benefited elite, White Southerners, these advantages did not necessarily cascade down to poor Whites in the same way that Northern economic prosperity did. Rather, Slavery resulted in widespread joblessness, economic insecurity, poverty, and illiteracy (Ash 1991) among Black Americans, as well as poor Whites (Helper 1857 : 377). [2] The scarcity of jobs and resources also reduced Whites' ability to secure competitive wages in the labor market (Helper 1857: 380) and played a part in wage depression more generally.

Following the abolition of slavery, efforts to undermine the economic and political incorporation of free Black Americans, in the form of Jim Crow laws, had similar implications for working class Whites. Although it is well understood that Jim Crow laws ensured legal residential segregation, political disenfranchisement, and the policing of the everyday lives of Black Southerners, the methods employed to achieve these goals also implicated many poor White Americans (Perman 2001:2; Valelly 2004:134). Despite a contemporary

[2] The economic prosperity of the North is apparent in its direct effects on individual wages. For example, cotton factories in Massachusetts paid workers 80 cents per day compared to 50 cents in Tennessee (Helper 1857: 379), indicating large wage differentials between regions.

understanding that the goal of Jim Crow laws was to ensure White supremacy, historical evidence questions this interpretation (Perman 2001:6). Jim Crow laws represented efforts by elite Whites to consolidate political power through voting rights restrictions, such as poll taxes, literacy tests, and arbitrary ballot procedures (Perman 2001) as well as the intentional fueling of racial tensions between lower-class White and Black populations (Gabriel et al. 2023) . Practically, these restrictions directly excluded Black and poor White populations from political participation, serving to fracture potential multiracial political coalitions (Perman 2001:2).

An illustrative example of these bi-racial impacts is evident in late 19th Century voting laws that sought to target Black voters, but in practice also suppressed poor White votes. For example, the Dortch Bill of 1889, a set of voting restrictions in Tennessee, implemented secret ballots in cities with large Black populations and banned assistance at voting booths for illiterate voters (Turney 1891) [3]. These measures restricted both illiterate Black voters and, unintentionally, large populations of illiterate Whites. This ultimately led to significant reductions in votes for both Democrats and Republican candidates in the South (Perman 2001: 54,59). In both Antebellum and Jim Crow periods, historical evidence indicates that, though not the modal White experience (Ash 1991), efforts and processes of racial stratification had scope for economic disadvantage and political disenfranchisement of Whites by virtue of their class position.

Despite evidence of the adverse effects of racial stratification on poor Whites, zero-sum views of race relations and racially coded issues (Gilens 1996) are pervasive, far-reaching, and have lasting impacts on social policy and political movements (McGhee 2021; Wetts and Willer 2018). Public opinion research shows that White Americans believe that they are the beneficiaries of systemic racism (Bobo and Zubrinsky 1996; Quillian

[3]The Dortch Bill, introduced by Tennessee Democrats in 1889, introduced a secret ballot meant to disadvantage illiterate voters by restricting assistance to illiterate voters. The sole purpose was to "benefit the rich man, burden the poor one and in the end result in the defeat of the Democratic party in the State,' (Perman 2001: 54). Quoting the Southwestern Christian Advocate (1891), Perman (2001) explains that in practice, this bill disenfranchised swaths of illiterate White voters. At the time, some Southern Democrats also objected to the Bill because of this loophole.The Dortch Bill was eventually ruled unconstitutional in Cook v. State of Tennessee (1891)

1995, 1996) and, by extension, view progress toward racial equality as coming at the expense of themselves. Black Americans also endorse similar zero-sum views of Black-White race relations, which starkly contrast coalition-oriented relationships observed among other racial-group pairings and emphasize the exceptional nature of Black-White relations (Warren and Valentino 2025). Explanations for why Whites hold zero-sum views are traced to perceptions of out-group threats to Whites' advantaged position in society (Blumer 1958; Wetts and Willer 2018). The presence of out-group threats, whether real or perceived, constitute challenges to their position (Blumer 1958) and provoke responses of racial resentment (Gilens 1996). These beliefs then have significant consequences for political efforts and support for non-racial redistributive policies that are characterized through a racial lens (Gilens 1996, 2000).

At the core of Whites' pervasive zero-sum beliefs about Black-White inequality is the assumption that processes of structural racism primarily advantage them. Zero-sum views can be understood as part of a larger set of compensatory and symbolic mechanisms that offset the consequences of racial stratification on working-class Whites. In Black Reconstruction in America, DuBois (1935) provides an illustrative framework that reconciles Whites' zero-sum beliefs with the deleterious consequences of racial stratification on working-class Whites. In the context of the post-Reconstruction U.S., DuBois details how White political elites forged a powerful alliance with the White working-class by promoting the salience and superiority of White identity that served to counteract potential Black-White class solidarity (DuBois 1935: 700) by casting *all* Whites as the beneficiaries of racial stratification.

> "It must be remembered that the [W]hite group of laborers, while they received a low wage, were compensated in part by a sort of public and psychological wage. They were given public deference and tides of courtesy because they were [W]hite," (DuBois 1935: 700). [4].

[4]"Whiteness" is intentionally capitalized in quotations where it was originally lowercase to correspond with the capitalization conventions of the rest of the manuscript.

Meanwhile, Black Americans were subject to mob violence, racial terror, and public insult. However, rather than racial subjugation resulting in the economic betterment of Whites, DuBois argues that the psychological wage paid to poor Whites, "was [such] that the wages of both classes could be kept low, the [W]hites fearing to be supplanted by Negro labor, the negros always being threatened by the substitution of [W]hite labor," (DuBois 1935: 701). Within DuBois' framework, zero-sum beliefs and racial divisions more broadly serve the symbolic purpose of overshadowing the shared class interests of working-class White, and Black populations by animating racial tensions.

In contemporary perspective, there are many parallels between racial segregation and earlier forms of racial stratification. Residential segregation is widely understood as the " 'structural linchpin' of race relations in America," (Bobo and Zubrinsky 1996: 883) that links racial stratification to unequal outcomes (Blokland 2008; Du Bois 1899; Light and Thomas 2019). Historically, Black-White segregation has been slow to decline and in the wake of Civil Rights legislation outlawing housing discrimination and legal segregation. Contemporary patterns of racial segregation are shaped, in part, by individual preferences and beliefs about racial integration and its consequences (Louie and DeAngelis 2024). Despite notable progress toward in the past several decades, a stubborn barrier to desegregation is White resistance to residential integration through White flight (Crowder and South 2008; Frey 1979; Logan et al. 2017), avoidance from cities (Rich n.d.), and suburbanization (Owens and Rich 2023). However, empirical evidence for the assumption that racial segregation leads to White advantage remains limited.

3 Theoretical Mechanisms Linking Racial Segregation to Longevity

To evaluate the extent to which segregation disadvantages Blacks and advantages Whites, I study longevity as a powerful indicator of individual life chances (Weber 1922). Longevity is both an important dimension of population health as well as an important axis of social inequality (Maralani and Portier 2021; Wrigley-Field 2024), including far-reaching implications for individuals' social networks, collective memory, culture, political participation, and social movements (Wrigley-Field 2025). Population health disparities between Black and White Americans are stark, in both contemporary and historical settings, and a key object of both scholarly and policy discussion. For example, in 2023, the difference in life expectancy at birth between White and Black Americans was of 4.4 years (Arias 2020). Longevity thus comprises more than a single health measure and reflects the cumulative summation of health inputs over the life course. There are a variety of contemporary mechanisms through which segregation may may affect longevity that unfold over the life course. While most segregation scholarship focuses on the downstream consequences of the stratified neighborhoods that Black and White individuals live in, segregation also affects county-level processes that affect all residents. In the following section, I demonstrate how (1) Neighborhood Effects and Spillovers, (2) Income Segregation, and (3) Political Context and Government Spending on public goods have significant potential to shape both Black and White longevity. In the following section, I outline these mechanisms, but formal tests are beyond the scope of the paper.

3.1 Neighborhood Effects and Spillovers

A leading pathway for segregation to affect longevity is through the differentiation of Black and White Americans' neighborhoods, leading to an uneven distribution of opportunities and risks. The neighborhood environments that Black American live in tend to have higher poverty (Massey and Denton 2003), elevated levels of crime (Light and Thomas 2019), greater exposure to hazardous conditions and pollutants (Wodtke, Ramaj, and Schachner 2022), and lower quality educational opportunities (Goldsmith 2009; Wodtke et al. 2023) than the neighborhoods of White Americans. Massey and Denton (2003) argued that increases in racial segregation create high concentrations of poverty, particularly for Black populations but also for Whites at smaller magnitudes (Massey and Eggers 1990). However, Ananat (2011) shows that segregation increased Black poverty while simultaneously reducing White poverty. Poverty also indirectly affects health through declines in employment opportunities (Wilson 1987) that strain individual resources (Aber et al. 1997). The timing and duration of poverty are also consequential particularly as they relate to early life adolescence (Brooks-Gunn and Duncan 1997; Chetty, Hendren, and Katz 2016; Sharkey 2013; Sharkey and Elwert 2011). Further, Massey et al. (2018) found that neighborhood poverty was associated with accelerated aging among Black and White mothers, indicating the compounding effects of exposure over the life course (DiPrete and Eirich 2006).

Another leading differentiator of Black and White neighborhoods is the concentration of crime and policing. Past work has established associations between segregation, homicide victimization, and aggressive policing (Bell 2020; Light and Thomas 2019). Exposure crime is associated with worse health outcomes (Foell et al. 2021) and aggressive policing elevates risk of homicide (Edwards, Lee, and Esposito 2019) in racialized ways. Empirical evidence shows that segregation increases Black homicide risk and reduces Whites' risk (Light and Thomas 2019). Therefore, the concentration of crime in predominately Black neighborhoods serves as a buffer reducing exposure in predominately White neighborhoods. Through this pathway, theoretical mechanisms and empirical evidence indicate that the consequences of segregation for crime exposure and their associated implications for health outcomes are racialized.

The residential separation of Black and White Americans can also affect resources and financial circumstances by increasing the cost-burden of housing in segregated places. Because income and socioeconomic status are foundational drivers of health status, housing-related cost burden could have substantial effects on longevity albeit through differing pathways for Black and White populations. Daniels (1975) found that White renters, intent on living in predominately White neighborhoods, paid a premium to segregate from Black populations.

Meanwhile, Akbar et al. (2025) showed that Black homeowners in pre-war U.S. cities paid a 28 percent premium to purchase a home in a majority White neighborhood arising out of a combination of cost and stigma. Housing cost burden could significantly affect longevity through reductions in spending power and potentially wealth accumulation (Hurd and Kapteyn 2003).

These neighborhood-level disparities contend a zero-sum relationship between segregation and health outcomes between Black and White Americans; wherein Black neighborhoods are exposed to higher disadvantage and White neighborhoods are shielded from it. However, the emphasis on neighborhood effects often overlooks the dynamic environments that individuals encounter through everyday, routine exposures. The variegated contexts that individuals navigate outside their neighborhoods provide significant scope for exposure to advantage and disadvantage that are not directly tied to residential neighborhoods (Browning et al. 2021; Cagney et al. 2020). Individual daily mobility patterns show that, rather than being isolated within residential neighborhoods, urban populations are mobile throughout the day. In any given day, individuals spend substantial amounts of time commuting to and from and at work and school, which expand the types of spaces they are exposed to. Browning et al. (2021) estimated that high school-aged youths spend about 34 percent of their waking time outside their residential neighborhoods. At the neighborhood-level, daily mobility flows are substantial, with inter-day mobility between neighborhoods being sufficient to radically reshape the demographics (Hess and Hall 2024). Given the profound role that non-neighborhood exposures play in everyday life, evidence indicates that activity spaces can expand individuals' exposure to poverty, crime, and disadvantage also shape health beyond ones' neighborhood. Sharp, Denney, and Kimbro (2015) found that the interaction between neighborhood and activity space context affects health: individuals of disadvantaged neighborhoods reported better health when they spent time in more advantaged neighborhoods and vice versa.

Activity space exposures also shape the dynamics of cross-group contact which matter for mental health and wellbeing (Blumer 1958). While interpersonal contact between Black and White individuals is reduced by segregation, both neighborhood isolation and interracial contact through non-neighborhood exposures can affect health. For example, Louie and DeAngelis (2024) argue that Whites' isolation from Blacks lead to negative racial stereotypes that shape assessments of neighborhood safety, danger, and crime. These negative stereotypes can lead to worse psychosocial distress and poor mental health for some Whites. Moreover, decreased contact also shapes the experiences of Black residents when they navigate predominately White spaces. Anderson (2022) describes a parallel set of processes, where Black individuals visiting White neighborhoods are subject to greater monitoring, scrutiny, stigma, and microaggressions from Whites, which have similar implications for mental health. Empirically, Hicken et al. (2014) found that Black Americans residing in Chicago reported higher levels of racism-related vigilance compared with Whites, which served as a mechanism for increased risk of chronic stress and hypertension.

3.2 Income Segregation

Another pathway, in addition to neighborhood-level mechanisms, are macro-level features of segregated places that shape the population health outcomes of all residents. One mechanism that is particularly consequential for individual outcomes, such as economic mobility, is income segregation (Chetty et al. 2014:34). Because income and race are tightly correlated (Massey and Eggers 1990; Reardon and Bischoff 2011) racial segregation is associated with income segregation, particularly among Whites (Jargowsky 1996).

A significant body of literature has drawn links between income segregation and downstream effects on long-run individual economic and educational outcomes that shape health. Chetty et al. (2014) found that income segregation significantly reduced the upward mobility of both Black and low-income White children (34). Moreover, Chyn et al. (2023) also found that racial segregation also decreased the upward mobility of Black and low-income White children. These effects on upward mobility are particularly consequential considering extensive research demonstrating the health returns to income and educational attainment (Montez and Bisesti 2024). Empirical evidence indicates that education increases longevity through a variety of economic and behavioral mechanisms (Halpern-Manners et al. 2020; Lleras-Muney, Price, and Yue 2022; Montez and Bisesti 2024). For instance, Halpern-Manners et al. (2020) found that an additional year of education resulted in .37 additional years of longevity for men born in the early 20th Century (1526). Moreover, Finch (2003) found that household income and material conditions were some of the most consequential determinants of infant health outcomes. A more limited body of work has also found direct links between income segregation and health. Mayer and Sarin (2005) found that economic segregation was associated with greater infant mortality. Similarly, Ross and Mirowsky (2001) found that income segregation was associated with higher working-age mortality.

3.3 Political Context and Government Spending

Segregation also may impact longevity through its influence on political context, government infrastructure, and portfolios of public policies across places (Montez et al. 2020). Prior literature has established that segregation is associated with lower spending on welfare, infrastructure, and public services (Ananat and Washington 2009; Chyn et al. 2023). Administrative units, such as the county or state, are important for health because they are intrinsically linked to important political, public health, legal, and institutional processes that affect individuals living within those boundaries. Places with more politically conservative policy portfolios regarding reproductive health, spending on education, income support, labor protections, and housing support (Montez et al. 2020: 672) were associated with reduced life expectancy among both men and women (Montez et al. 2020: 668). More generally, ethnographic work has provided support for these patterns, showing that low-income Whites, especially in rural areas, are particularly vulnerable to the health effects of conservative policy portfolios (Metzl 2019).

Beyond policy decisions that concern health, racial segregation also shapes public support and efficacy of progressive policies related to employment, labor relations, and redistributive policy. Ananat and Washington (2009) found that segregation reduced the efficacy of working-class oriented policies that would improve the economic position of low-income residents. Chyn et al. (2023) also found that racial segregation produced higher levels of racial resentment among White Americans which may serve as a mechanism to decrease electoral support for progressive policy. Whites' racialized attitudes have long been shown to have implications for views about welfar and redistributive policy, which have direct implications for electoral outcomes (Gilens 1996; Wetts and Willer 2018). Indeed, Chyn et al. (2023) also found that racial segregation was associated with decreases in spending on social welfare, public goods, and infrastructure. Decreases in funding for social welfare programs providing income and nutritional support have particularly strong implications for the health and wellbeing of low-income individuals reliant on these services regardless of race.

Moreover, a key dimension of government infrastructure is funding for public health services that have contributed to significant contemporary and historical improvements in population health (Cutler and Miller 2005; Liang et al. 2023). Reductions in spending on public health services and infrastructure thus have important consequences for population health and disparities more generally, regardless of residence within an area. Through this mechanism, variation in county-level health services and infrastructure could significantly shape longevity.

4 Prior Work on Racial Segregation and Health

Past work has established that racial segregation is associated with negative health outcomes for Black Americans and wide Black-White health disparities. While most studies focus on the health outcomes of

Black Americans, a limited number of studies consider the effects of segregation on White Americans, but with mixed results due to methodological limitations.

One strand of research has drawn links between racial segregation and infant birthweight. Austin, Harper, and Strumpf (2016) found that segregation decreased the birthweight of Black children by 2.8 grams and decreased the birthweight of White children by 0.63 grams. However, another study by Vu, Green, and Swan (2024) found negative impacts of segregation on the birthweight of Black children and no effect on the birthweight of White children.

On mortality, Logan and Parman (2018) found that, in certain contexts, segregation was associated with greater mortality risk for White Americans and lower mortality risk for Black Americans. Similarly, Karbeah and Hacker (2023) found that segregation was associated with a greater risk of mortality among White and Black children. However, the generalizability of the conclusions from Logan and Parman (2018) are limited because they rely exclusively on death records from North Carolina. Moreover, Karbeah and Hacker (2023) use a measure of segregation that is incongruent with established segregation theory and measurement, because of data limitations. Instead, their measure reflects processes at the neighborhood-level, because their measure of segregation is based on the racial composition of one's neighbors, rather than the distribution of groups within a place. Because this measure reflects a different set of processes, which the authors acknowledge, than what is conceptualized by contemporary segregation scholarship, it is unclear how to understand these results in the context of established measures of segregation.

While existing studies provide evidence of associations between segregation and poor health outcomes, particularly for Black Americans, results across studies for White Americans are mixed and are largely correlational. Demographic work has demonstrated that there are many aspects of places, such as the policy context, infrastructure, and environment, that are consequential for population health that are related to segregation (Montez et al. 2020). Currently, these results provide suggestive evidence but are not able to establish causality. Therefore, if residential segregation and longevity are shaped by unobserved heterogeneity or confounded by other features of counties, then correlational analyses cannot distinguish whether relationships are causal.

Finally, it is important to distinguish the relationship between segregation and health and health disparities. Using life table methods, Hendi (2024) showed that segregation was responsible for a 16-year increase in the Black-White life expectancy gap for men and a 5-year increase for women between 1990-2019. While studies of Black-White longevity gaps are informative for characterizing population-level inequalities, they obscure important information about the direction of these effects and are substantively ambiguous without further context (Wrigley-Field 2025). If segregation positively affects White health and Black health negatively, or affects both groups negatively, then these relationships both produce inequality. Distinguishing between these relationships is important, because longevity inequalities that arise from the same or different signed relationships point toward distinct implications for sociological theory.

4.1 The Present Study

The goal of this paper is to estimate the plausibly causal effect of segregation on the longevity of Black and White Americans born to early 20th Century birth cohorts and residing in U.S. counties between 1980-2000 in later life. To establish the effects of segregation on longevity, I rely on a validated instrumental variable approach that leverages variation in local government fragmentation within U.S. counties in 1967. This instrument induces quasi-random variation in racial segregation, through a Tiebout mechanism. A Tiebout mechanism is the tendency of individuals to "vote with their feet". In areas with more local governments, there is a greater variation in residential options in the form of differing tax rates and services, which allows individuals to sort according to their preferences. Greater sorting leads to higher segregation because households with similar backgrounds and resources cluster in similar areas, but does not necessarily affect individual outcomes. I combine this quasi-random variation with individual-level full count Census data linked to Social Security Numident mortality records. In accordance with historical record and prior scholarship, I also examine how the effects of segregation on longevity vary by educational status to distinguish whether educational status moderates the effects of segregation on longevity. Establishing the causal effects of segregation on Black and White longevity contributes important clarity about whether Whites' zero-sum beliefs reflect realized benefits from segregation or symbolic benefits that serve as a mechanism to offset the effects of racial stratification.

5 Data

5.1 Sample and 1940 Linked-Census Data

The empirical analysis of this paper is anchored by individual-level demographic data records from the 1940 full fount Census (Ruggles et al. 2023). To construct my analytic sample, I identify individuals from the 1905-1920 birth cohorts who were 20-35 in 1940. These individuals are then linked to Social Security Administration (SSA) mortality records between 1988 and 2005 provided by the Berkeley CenSoc Numident file (Breen, Osborne, and Goldstein 2023). These data offer unparalleled individual-level data on mortality that are geolocated to contemporary residence at death and which are representative of the population living in the United States in 1940. CenSoc Numident records cover deaths occurring between 1988 and 2005, allowing for reliable estimates of longevity conditional on an individual surviving to age 65 (Goldstein et al. 2023)[5]. The geographic specificity and rich covariates make these data ideal for analyzing mortality and longevity disparities.

[5]The interpretation of longevity conditional on surviving to 65 is an artifact of using truncated mortality data which only reliably cover deaths over the age of 65 (Breen et al. 2023: 35) and is interpretation used in (Breen 2024).

I focus on the 1905-1920 birth cohorts for two reasons. First, their birth years and the end year of the mortality observation window allow for plausibly complete observation of mortality that is not biased by right censorship (i.e. missing the oldest old). In other words, those born in 1905 would be 100 years old in 2005, making it reasonable to assume that few individuals remain alive. Even for those individuals who live very long, using doubly truncated data will result in estimates biased toward zero (Breen 2024), and thus produce estimates of the effect of segregation on longevity that are likely conservative. Second, restricting to ages 20-35[6] allows for confidence that individual-level confounders to longevity, such as education status, are observed. By conditioning on those 20 years or older, it is reasonable to assume that individuals have completed the modal level of formal education by the time that they appear in the sample.

It is well known that the Numident data suffer from the problem of double truncation, where data are truncated on both the left and right side. Goldstein et al. (2023) show that when not accounted for, doubly truncated mortality observation windows result in biased estimates of longevity disparities toward 0 and higher mean observed death age in samples. This means that those born in 1905 are observed between the ages of 83-100 years old and those born in 1920 are observed between the ages of 68-85 years old. Birth year fixed effects are integral to reducing bias in mortality analysis with truncated data (Goldstein et al. 2023), where individuals of varying birth cohorts may have differing numbers of years covered by the death window.

There are also representativeness features of the data that bear on substantive interpretations. The CenSoc Numident file links 1940 Census data to the public release of the National Archives Record Administration "NARA Numident" file using the Abarmitzky, Boustan, and Eriksson (ABE) exact matching algorithm (Abramitzky, Boustan, and Eriksson 2012). For Black Men and Women, comparisons with the Census indicate that the Numident file slightly underrepresents these groups (Breen et al. 2023).[7] To add further validity to

[6]Halpern-Manners et al. (2020) finds that the mean number of years of education of decedents in the 1940 Census is 10.4 years (1525). If we assume a normative time of starting school at 6, individuals in the sample reach the average at 16.4 years, which is well before our age cutoff.

[7]For Black Men the Numident links are slightly underrepresented at a -1.8 percent difference from the Census and slightly

these estimates and address representation issues, I use post-stratification weights developed by Osborne (2023) that re-weight CenSoc Numident counts to period-specific distributions of deaths to adjust for possible selection. Table 1 describes the representativeness of the population.

5.2 County-level Data

I examine exposure to segregation in mid-to-late life, obtained and calculated based on the county individuals resided in at death. Examining the county at death offers an empirical understanding of the effects of segregation that occurred after the Great Migration and the passage of Civil Rights Act that outlawed legal segregation. Because much of the contemporary segregation scholarship examines the consequences of de jure segregation, this time period aligns more consistently with these theories. Scholars recognize that the geographic scope and structure of segregation after the Great Migration are understood to be distinct from the segregation that occurred before, predominately in the South (Grigoryeva and Ruef 2015; Leibbrand et al. 2020).

Despite potential limitations of using one's county at death, it is reasonable to believe that place characteristics are highly correlated with their residential experiences at birth. Patterns of neighborhood attainment and residential mobility over the life course show that the modal neighborhood transitions are to places with similar characteristics (Sampson and Sharkey 2008). Moreover, across the life-course, migration propensity declines with age (Bernard, Bell, and Charles-Edwards 2014) with individuals less likely to migrate after age 40 old and, when they do, it is usually short distances (Molloy, Smith, and Wozniak 2011; Zimran 2022). For these reasons, segregation in the county of death likely represents a period of exposure encompassing mid-life and is correlated with earlier experiences. However, to account for any residual variation due to early life exposures and the exceptional nature of moves during the Great Migration, I include birth county fixed

more at a -2.4 percent difference (Breen et al. 2023: 8-9). This matching procedure may be particularly selective for unmarried women, because of changes of marriage-related name changes over the life course. Fortunately the Numident data contain parents last names that can be used to link women who have undergone name change.

effects in all models.

To measure segregation, I use the dissimilarity index (D) (Duncan and Duncan 1955). The dissimilarity index measures the evenness of the distribution of Black and White residents within a county, ranging from 0-100, and is interpreted as the share of Black or White individuals that would have to move to a different neighborhood for the racial composition of neighborhoods to be the same as the racial composition of the county. The dissimilarity index is widely used in prior work (Cutler and Glaeser 1997; Hendi 2024; Light and Thomas 2019; Massey and Denton 2003) because it is straightforward to calculate, concordant with past empirical evidence, and has a concrete interpretation. Nevertheless, the D is well known to suffer from various index biases, outlined in Fossett (2017) and others (Carrington and Troske 1997; Winship 1977), particularly as they relate to temporal comparisons and variability in the population composition and relative group size of counties. Accordingly, in supplemental analyses, I replicate the main results with a bias-adjusted D index that adjusts for this bias (Fossett 2017)[8].The dissimilarity index for the county an individual resided in at the time of their death is calculated formally in Equation (1):

$$D = \frac{1}{2}\sum_{i}^{N} |\frac{b_i}{B} - \frac{w_i}{W}| \times 100 \tag{1}$$

Where N is the number of counties, b_i and w_i are the proportion of Black and White residents in the ith census tract and B and W are the proportion of Black and White residents in the county. The dissimilarity scores are calculated from decennial Censuses 1980, 1990 and 2000, which correspond to the year of death observed for sample members.

Counties are additionally linked to 1967 Census of Governments data on the number of municipal and

[8]In contrast to D, the bias-adjusted D measure is more complex to estimate and the interpretation less concrete (Fossett 2017: 233). For these reasons, I use the conventional dissimilarity index for main analyses.

township governments and their federal revenue share from the Government Finance Database (Pierson, Hand, and Thompson 2015) that I use for identification. My final analytic sample consists of ($N = 2,094,017$) individuals residing in $1,774$ counties at their time of death over three decades from 1980-2000. The resulting counties are filtered from all counties in the U.S. such that they have a Black population greater than 0 and are not missing data in 1967 on the county finance measures used as instruments [9].Table 1 displays the individual and county characteristics between those in the 1905-1920 birth cohorts in and out of the sample and demonstrates that the differences are minimal.

6 Methods

6.1 Empirical Strategy

There are many features of counties and places that also shape health (Montez et al. 2020), leading associations between county-level segregation and longevity to be biased. A leading strategy employed in prior scholarship to address concerns of endogeneity are instrumental variables that create exogenous variation in segregation but do not plausibly affect health. I use a validated instrument first used in Cutler and Glaeser (1997) and employed subsequently by Light and Thomas (2019) that leverages variation in the levels of government fragmentation of counties in 1967 that induce geographic sorting between neighborhoods. I instrument for segregation using (1) the number of municipal and township governments within a county and (2) the share of state intergovernmental revenue allocated to these local governments. The intuition behind this approach is that areas with more local governments offer residents more opportunities to "vote with their feet" in selecting among jurisdictions that differ in taxes, services, and demographics. This greater potential for

[9]Beyond filtering to counties that have non-zero Black populations, the sample size is further reduced by variability across years in Census of Government reporting rates. Pierson et al. (2015) explain that annual sample sizes of governments vary widely, which result in missing data, and reporting at smaller government levels, such as the township and municipality, vary in size and quality year-to-year (22). However, years ending in 2 and 7 are the most representative, making 1967 an ideal candidate year because it is the earliest year that data is available and of the highest relative quality in terms of data coverage (Pierson et al. 2015 :22).

sorting fosters more residential segregation, as households with similar preferences and resources cluster within the same localities.

The use of municipal and township governments to instrument for contemporary population patterns are appealing for multiple reasons. First, as demonstrated by Cutler and Glaeser (1997) and Light and Thomas (2019), the number of governments over time is largely constant, with a correlation over time of .98 (Cutler and Glaeser 1997:848). Certain government patterns are fixed stretching as far back to 1787, suggesting that government placement is not affected by contemporary sorting patterns. Second, because of their constant nature, the number of governments are likely to result in spatial sorting that is informed by quasi-random early settlement geographic features, such as the arrangement of hills, rivers, and other geologic features that affect segregation patterns but are unrelated to contemporary demographic forces that offset segregation and health outcomes (Cutler and Glaeser 1997). An implication of this constant nature is that, as segregation has declined over time, the predictive association between governments and segregation should also decrease over time. I consider the impact of these over-time changes in the discussion of IV assumptions.

In analyses, I include the log of the number of governments and the state-level share of intergovernmental revenue to purge local endogenous factors. When evaluating the plausibility and relevance of an instrument, it is important to do so in the context of IV assumptions (Felton and Stewart 2024): (1) the instrument is exogenous and relevant; (2) there is a strong first stage relationship; and (3) monotonicity or the instrument only influences the level of segregation in a county in one direction.

The first assumption is that these instruments impact longevity only through segregation. To further validate this assumption, I follow Cutler and Glaeser (1997) and adjust for inter-governmental revenue shares which have the advantage of accounting for endogenous variation in resources at the local government level that may also affect longevity [10].

It is also important to recall that the instrument is the density of governments in a county and does not rely on any government-specific quality metrics. Second, in Figure 1 I demonstrate that association between the instruments and segregation at the county-level is highly predictive using a binned scatter plot. Tables of the main IV models also show the first stage for individual-level regressions.

[Figure 1 here]

The first-stage F-statistic assesses the strength of the instrument and segregation and indicate a strong and significant F-test of 226.53. This test gives confidence in the assumption that governments strongly predict racial segregation and are not subject to bias that may result from weak instruments (Angrist and Kolesár 2021). Third, monotonicity requires that the instrument pushes segregation in the same direction for all units. While a direct test of this is not possible, it is possible to examine patterns consistent with monotonicity to provide evidence in support of this assumption. Accordingly, I employ a similar test to Rauscher (2020) to examine demographic patterns that would be consistent with monotonicity. It is well known that segregation levels have declined over time. If we believe that part of this decline is due to less economic and racial residential sorting, then we would also expect that year-over-year the association between the instruments and segregation would become weaker. One way to do this is to examine how the strength of the association between the instruments and segregation changes decade-over-decade. Consistent with this

[10] Because there are more instruments than endogenous regressors, a natural test of the validity of many instruments are overidentification tests. These tests are commonly interpreted as testing the" validity" of the instruments, however overidentification tests must first assume that one instrument is valid (e.g., in the just-identified case) (Felton and Stewart 2024), which is not informative for assessing validity. A more informative interpretation from Mogstad et al. (2021), is that rejection of overidentification tests suggest that there is heterogeneity in treatment effects across subpopulations affected by the different instruments, which is not in reference to validity. Additionally, the overidentification tests assume constant treatment effects, which is unreasonable in a setting where the IV estimate is the local average treatment effect. Using the same instruments, Cutler and Glaeser (1997:854) fail to pass overidentification tests for the instruments because of large sample sizes ranging from 3-6 thousand observations. As expected, the instruments generally fail to pass these tests due to the large sample size of my data where observations are in the millions. An alternative to overidentification tests is placebo analyses that require that a researcher find an additional instrument that shares all the same confounders as the original instrument but is thought to not affect the treatment (Felton and Stewart 2024). While informative, the shared confounders assumption of placebo instruments is also quite strong and the search for a candidate placebo instrument faces the same challenges as credible instrument.

assumption, the strength of the association between the instruments and segregation, as measured by R^2 decreases in size between 1980-2000. While this does not represent a conclusive test of monotonicity, these patterns are consistent with the assumption.

Conditional on these assumptions, IV estimates of a regression of the effects of segregation on longevity are interpreted as the local average treatment effect (LATE) (Imbens and Angrist 1994; Mogstad, Torgovitsky, and Walters 2021), or the treatment effect amongst those living in counties where segregation increased because of greater government density. These instruments have been used effectively in prior studies to examine how segregation effects multiple outcomes (Cutler and Glaeser 1997; Light and Thomas 2019).

6.2 Estimation

Empirically, I estimate the relationship between segregation and age at death, conditional on living to age 65, in the county of residence at death. I estimate the model via two-stage least squares estimation, stratified by racial group. I control for a rich set of individual and contextual covariates to adjust for any residual confounding. In fully specified models, I control for sex, education, marital status, employment status, whether an individual migrated, whether they reside in the South, urban-rural fixed effects [11], birth year and state fixed effects, and occupation fixed effects. In all models, I include robust standard errors to account for heteroskedasticity. Equation (2) describes the model[12]:

$$Y_i = \beta_0 + \beta Segregation_i + \boldsymbol{\beta} X_i + \gamma_{b(i)} + \lambda_{c(i)} + \epsilon_i \tag{2}$$

[11]Between 1993 and 2003, Rural-Urban Continuum codes changed from their definitions between 1983-1993. For 1983-93 codes, metro counties were classified using a scheme ranging from 0-3, with counties that belonged to a metro area with a population of 1 million or more were categorized into Central and Fringe counties. In 2003, this scheme was revised so that counties were classified based only by population, with Fringe and Central collapsed into one category. Accordingly, to increase comparability. I collapse codes 0 and 1 in the 1983 and 1993 continuum code schemes into one category so that they are harmonized 1-3 scheme used in 2003.

[12]I also re-run models with standard errors clustered at the county-level the results are robust to alternative standard error specifications.

The primary goal is to estimate $\beta segregation$, which is our focal measure of segregation operationalized as the dissimilarity in an individual's county at death. $\boldsymbol{\beta}$ is a vector of regression coefficients and X is a vector of individual-level control variables and γ and λ are fixed effects for individual birth year and birth county.

In summary, a core strength of this approach is that the instruments address critical endogeneity and selection challenges to estimating the relationship between segregation and longevity and rich individual-level covariates address additional confounding. The Results section proceeds as follows. The first part describes the characteristics of individuals in my sample and presenting results of the OLS association between segregation and longevity. The second part presents the instrumental variable results, heterogeneity by educational status, and a set of robustness tests.

7 Results

7.1 Main Results

First, I present descriptive statistics of the analytic sample (N = 2.1 million) compared to the full sample in Table 1 and by racial group for the analytic sample in Table 2. Between samples, the analytic sample appears to be broadly representative. However, analytic sample members live in counties that have larger populations, are slightly less likely to live in the South, and are more educated than those in the full Numident sample. By racial group (Table 2), White Americans have a clear longevity advantage over Black Americans and are more educated. Black Americans are significantly more likely to live in the South and live in counties with a significantly higher Black populations that are more segregated than White Americans.

Next, I turn to the results from OLS regression models of the associations between segregation and longevity. Figure 2 demonstrates that segregation is associated with reductions in the longevity of both Black and White Americans with Black Americans facing a substantial mortality penalty relative to White Americans.

A 10-point increase in dissimilarity is associated with a 0.84-year decrease in years of life for Black Americans and a 0.72-year decrease for White Americans (Table 3). These associations are also robust to the inclusion of covariates and fixed effects in the adjusted models.

[Figure 2 here]

Next, I turn to the results from instrumental variable regression models estimated via two-stage least squares. Figure 3 presents instrumental variable estimates of the effect of segregation and longevity. These estimates confirm the OLS finding that segregation reduced the longevity of Black Americans and White Americans but with a notable reduction in magnitude. The unadjusted models indicate that a 10-point increase in segregation reduces Black longevity by 0.40 years and White longevity by 0.36 years. Adjusting for covariates increases the magnitude of the effect of segregation for Black Americans to 0.44 years and attenuates the association for White Americans further to 0.29 years. These results indicate that racial segregation substantially reduces the longevity of Black Americans and modestly of White Americans. In comparison to OLS associations, IV results suggest that the OLS significantly overstates the magnitude of effects for Black and especially, White Americans. The difference in the respective magnitudes of the OLS and IV coefficients is likely due to unobserved heterogeneity at the county and individual level.

[Figure 3 here]

In addition to main estimates, I also stratify IV models by gender to understand the degree to which gender and race intersections may matter for the effects of segregation. Figure 4 shows that the effects of segregation on longevity for Black men and women are virtually indistinguishable (diff = 0.02), but there is a slightly larger reduction in the harmful effects of segregation for White women's longevity compared to White men.

[Figure 4 here]

A 10-point increase in segregation corresponds to a 0.31-year decrease in longevity for White men compared to a 0.28-year decrease for women – a difference of 0.03 years. In combination with the main results, results by gender indicate that there are inequalities among Whites by gender that advantage White women compare do men, however the mechanisms leading to this difference beyond well-known gender differences in mortality are beyond the scope of this paper.

[Figure 5 here]

Next, I address a supporting research question that examines how the effect of racial segregation on longevity varies by educational status. Recalling predictions from DuBois's wages of Whiteness theory, if Whites are differentially harmed by racial stratification, then it would be expected that the effects of segregation on longevity vary by educational status. Figure 5 and 6 display the results from the IV models of racial segregation on longevity interacted with education level for White and Black Americans. First, results for Whites reveal that education is a significant moderator of the effects of segregation on longevity, such that the most advantaged Whites are harmed less by segregation.

For those with a college degree or higher, segregation reduced White longevity by 0.19 years. In contrast, those with some college or less have the largest longevity penalties. These results underscore a clear class gradient to these effects, suggesting that higher education buffers the negative effects of segregation on Whites. In contrast, Figure 6 demonstrates a remarkably consistent effect of segregation for Black Americans, regardless of education. Despite variability in the coefficients by education level, statistical tests of the interaction and differences between coefficients show that even at higher levels of education, segregation substantially reduces Black longevity.

[Figure 6 here]

7.2 Robustness Checks

As a robustness exercise, I examine results using post-stratification weights to adjust for potential selection relative to period-specific death rates, and by using the Theil H index (Iceland 2004), Isolation index, and bias-corrected Dissimilarity Index as alternative measures of segregation.

First, I re-estimate IV regressions using CenSoc post-stratification weights that adjust for lingering representativeness issues. Results from weighted regression models increase the magnitude of effects for both White and Black Americans with and without controls compared to unweighted results and underscore the same relationships as the main results.

Second, I examine the main IV results with two alternative measures of segregation: the entropy-based Theil index and the isolation index, which measure the probability that Black and White individuals interact within a geographic parcel. Using the isolation index, Theil H index, and bias-adjusted D, these alternative measures of Black-White segregation resoundingly confirm the main empirical results using the dissimilarity index. For Black results specifically, the H index attenuates the estimated effect slightly, but the sign and significance of the effect is unchanged. The bias-adjusted D also confirms these results. Overall, across different subgroups, specifications, and measures, the results are quite consistent. Racial segregation substantially decreases the longevity of Black Americans and modestly decreases the longevity of White Americans.

8 Discussion and Conclusion

Guided by zero-sum theories of processes of racial stratification, the prevailing expectation is that segregation increases White longevity and reduces Black longevity. Though past work finds associations between racial segregation and Black-White health disparities, correlational studies are limited by endogeneity and unobserved heterogeneity that cannot be interpreted as causal. Despite expectations that segregation unambiguously advantages Whites, historical and demographic evidence indicate there is significant scope for processes of racial stratification to have unintended effects on lower-class White Americans. In this paper, I test the argument that White Americans necessarily benefit from racial segregation using linked 1940 full count Census data and an empirical strategy leveraging variation in government fragmentation of counties.

In naive OLS regression models, IV models that address selection and endogeneity concerns, a variety of alternative specifications, and across alternative measures of segregation, the empirical results are highly consistent. Contrary to zero-sum expectations, racial segregation substantially and significantly reduces Black longevity and modestly reduces White longevity. It would be inaccurate to claim that residential segregation harms Black and White Americans in the same ways; rather, segregation reduced Black longevity at a magnitude approximately 50 percent greater than that of White Americans. At the same time, these results reject straightforward zero-sum assumptions of White advantage. Though modest, these effects suggest that desegregation may positively benefit White Americans, rather than adversely affecting their life chances. In the context of noted declines in racial segregation between the 1970s to 2000s of approximately 21 points [13], a shift from 1970 to 2000 average county racial segregation corresponds to a gain of 11.09 months of life for Black residents and 7.31 months for White residents. In the context of other studies concerning longevity, these results are substantial. For instance, using the 1940 Full Count data, Breen (2024) estimates that homeownership increased longevity by .31 years (or 4 months) and Halpern-Manners et al. (2020) estimate that each *additional* year of education increased longevity by .35 years (or 4.2 months). These results speak to the far-reaching impact of racial stratification for the lives of Black and White Americans (Louie and DeAngelis 2024; McGhee 2021).

In connection with the main results, it is important to emphasize that the effects of segregation are

[13]The 21-point difference is derived from the author's calculation of the difference between the average levels of segregation in sample counties in 1970 and the average in 2000.

differentiated by educational status for White Americans but not for Black Americans. For White Americans, attaining a college degree or higher is associated with modest reductions in the harmful effects of segregation on longevity, however this does not entirely offset longevity reductions. Importantly, results overall contest zero-sum expectations, but reveal important heterogeneity in the effects of segregation on longevity, with highly educated Whites affected the least. By contrast, for Black Americans, the effects of segregation are equally harmful regardless of education level. The inequality in the protective effects of education by race is consistent with a growing literature on differential health returns (DeAngelis, Hargrove, and Hummer 2022) that finds that Black Americans, even at higher status levels, do not reap the same health benefits as similarly advantaged White peers. More generally, these these findings illuminate how, by virtue of class position, segregation is especially harmful for lower-educated White Americans.

This paper makes several important contributions to research on the consequences of segregation as well as the sociology of race and race relations. Prevailing views on racial stratification, of which segregation is a key mechanism, have focused on how segregation creates divergent outcomes between Black and White residents, with Blacks suffering the harmful ramifications and Whites reaping the benefits. Two past studies, Light and Thomas (2019) and Quillian (2014), test the hypothesis that Whites benefit from segregation in terms of homicide victimization and educational attainment. While Light and Thomas (2019) finds support for a zero-sum relationship between segregation and crime, Quillian (2014) finds that segregation decreases Black educational attainment but is ambiguous for Whites. Despite the salience of the Black and White neighborhoods for outcomes, these perspectives overlook the macro-level mechanisms of segregation, particularly the role that population processes play. As I argue in the Mechanisms Section, segregation results in a variety of population-level externalities related to public opinion, government, and economic segregation that shape the experiences of all residents. This paper contributes to sociological theory more generally, by highlighting the importance of considering how macro-level processes and individual class position intersect with mechanisms of racial stratification.

In accordance with theories developed by DuBois (1935), the wages of Whiteness serve a symbolic role in offsetting the pernicious, class-based effects of racial stratification on lower-status White Americans. Zero-sum expectations that Whites are better off in a racially stratified society and disadvantaged by desegregation can be understood as serving a similar purpose. Concordant with theory and historical record, these findings show that the harms of segregation are particularly severe for low-educated Whites. However, while DuBois' theory argues that these effects are felt primarily by lower-class Whites, these findings show that the effects extend to even the most advantaged Whites. Like earlier efforts to disenfranchise Black Americans under Jim Crow, I these results demonstrate how the effects of racial segregation occur both along racial, but also class-lines.

This study is also not without limitations. First, there is inherent selectivity embedded in examining years of life conditional on surviving to 65. While this analysis is informative for understanding longevity in this period, it cannot speak to inequalities that may arise in early and mid-life that may differentiate Black and White populations. However, because these results reveal substantial effects of segregation on longevity even in this selective population, the disparities observed within the full population may be larger. Moreover, the doubly-truncated nature of these data results in regression estimates that are biased toward zero, suggesting that these estimates are conservative.

Second, while instrumental variables are powerful causal inference tools to address selection in cross-sectional settings, they require strong assumptions that cannot directly be tested. I demonstrate that the relevance condition is strongly supported and descriptive evidence is consistent with monotonicity. Relying on prior research and reasoning, the historical nature of the instruments situated well before the study is suggestive that exclusion is supported, but the exclusion restriction cannot be directly tested. Despite these limitations, and differences in the magnitudes of the coefficients, the conclusions from OLS and IV regression models are the same.

Third, while this analysis leverages historical birth cohorts, these results cannot speak to more contemporary relationships. Because segregation has declined over time, it is reasonable to believe that effects for more recent birth cohorts may be attenuated compared to this population. Nevertheless, it is important for future research to consider how the effects of segregation on health have changed over time.

These limitations provide promising suggestions for future research. Beyond using full count census data and information on completed mortality, future research could leverage representative surveys and information on health conditions that are predictive of mortality. For instance, examining outcomes like Allostatic Load could allow researchers to gain a contemporary understanding of the ways that segregation shapes processes relevant to mortality. While the study of mechanisms is beyond the scope of this study, I have outlined several compelling population and neighborhood-level mechanisms of segregation that plausibly affect individual outcomes. Future work could consider studying these mechanisms directly to determine the primary mechanisms through which segregation affects individual outcomes and whether mechanisms vary by the outcome of interest.

If there is one takeaway from this paper, it is that the processes of racial stratification have severe consequences for the lives of Black and White Americans. In the context of historically fraught Black–White relations in the United States, Whites' persistent resistance to racial integration remains a barrier to desegregation. A central component of this resistance is the misinformed belief that racial segregation necessarily advantages Whites. Although zero-sum understandings of racial inequality are widespread, this study shows that segregation also harms Whites. While those with greater education are buffered from the harmful effects of segregation, those advantages are modest. For the vast majority of Whites, segregation negatively affects life chances by reducing longevity. Crucially, this paper demonstrates that processes of racial stratification intersect with population processes and individuals' class position to shape the lives of both Black and White Americans in ways that are counterintuitive. In line with DuBois (1935), this paper provides empirical support for the Wages of Whiteness argument: casting Whites as unequivocal beneficiaries of segregation serves as a symbolic mechanism that obscures the class-based harms of racial segregation. These findings suggest that policies promoting racial integration could improve health and life chances for both groups and narrow Black–White disparities.

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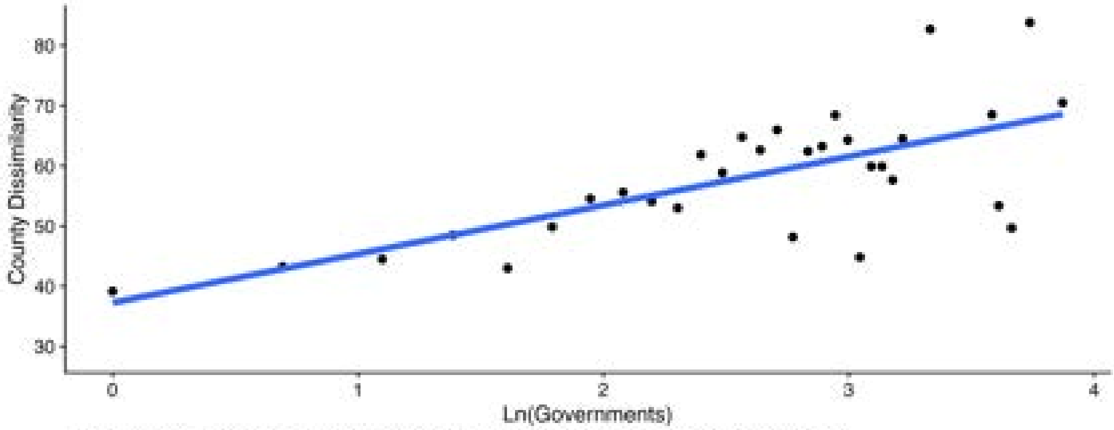
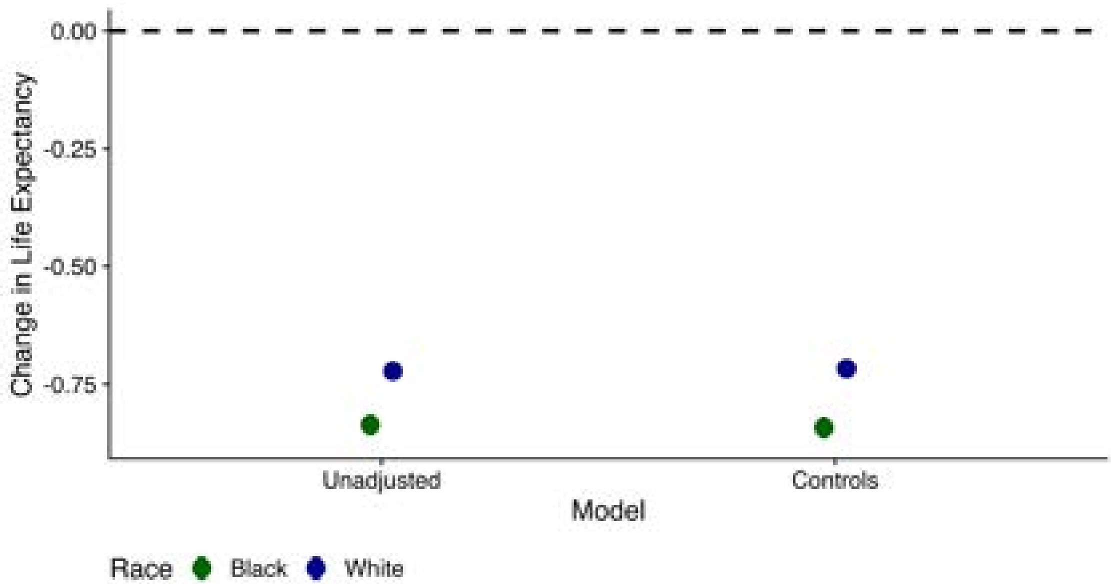
10 Tables and Figures



This figure displays the association between government fragmentation and segregation. Segregation is measured by the index if dissimilarity that measures evenly distributed Black and White residents are within a county. Dots represent means of bins at each level of the instrument. The blue line corresponds to the fitted OLS regression line of segregation on the instruments. Ln(Governments) refers to the log of the number of governments: the focal measure of government fragmentation. F-statistic = 236.53.

Figure 1: First-Stage Relationship



This figure displays estimates of the OLS association between racial segregation and longevity. The unadjusted model contains birth year, urban-rural, and birth state fixed effects. The model with controls adds covariates and occupation fixed effects. Estimates refer to a 10 point increase in Dissimilarity.

Figure 2: OLS Estimates of the Association Between Segregation and Longevity

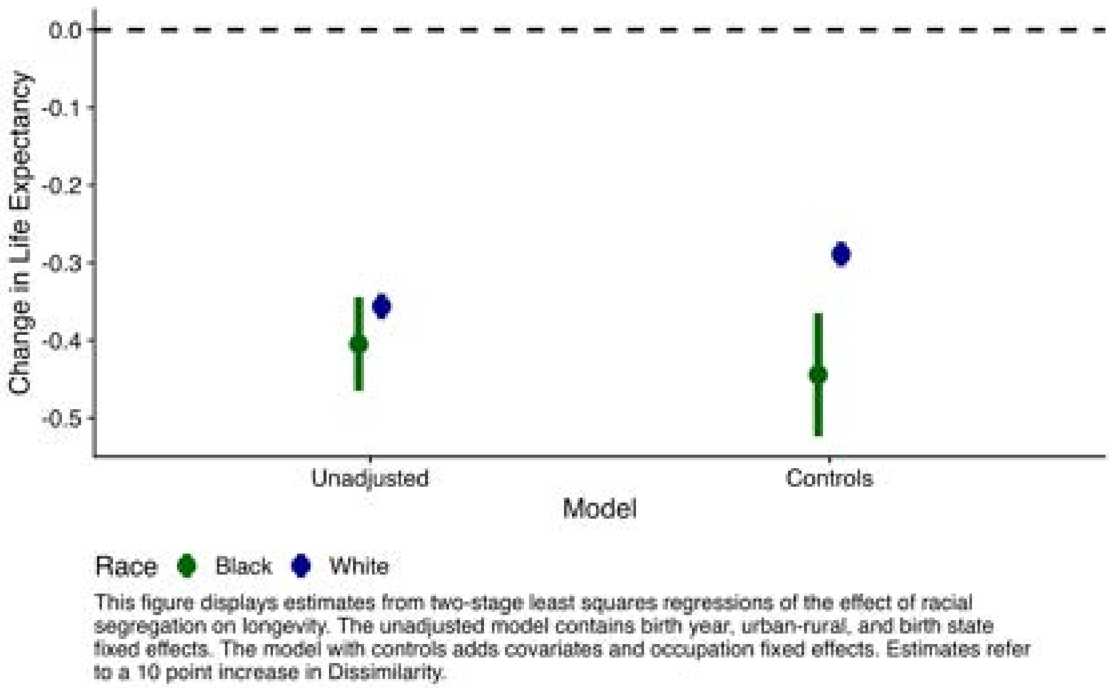


Figure 3: IV Estimates of the Effect of Segregation on Longevity

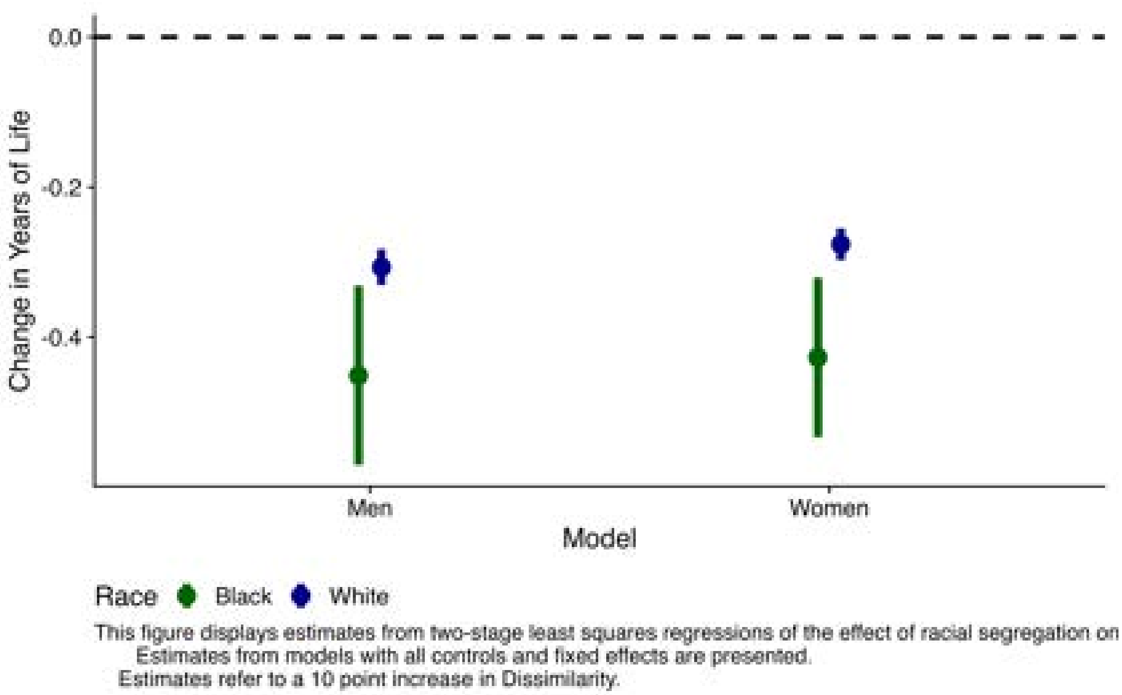


Figure 4: IV Estimates of the Effect of Segregation on Longevity (By Gender)

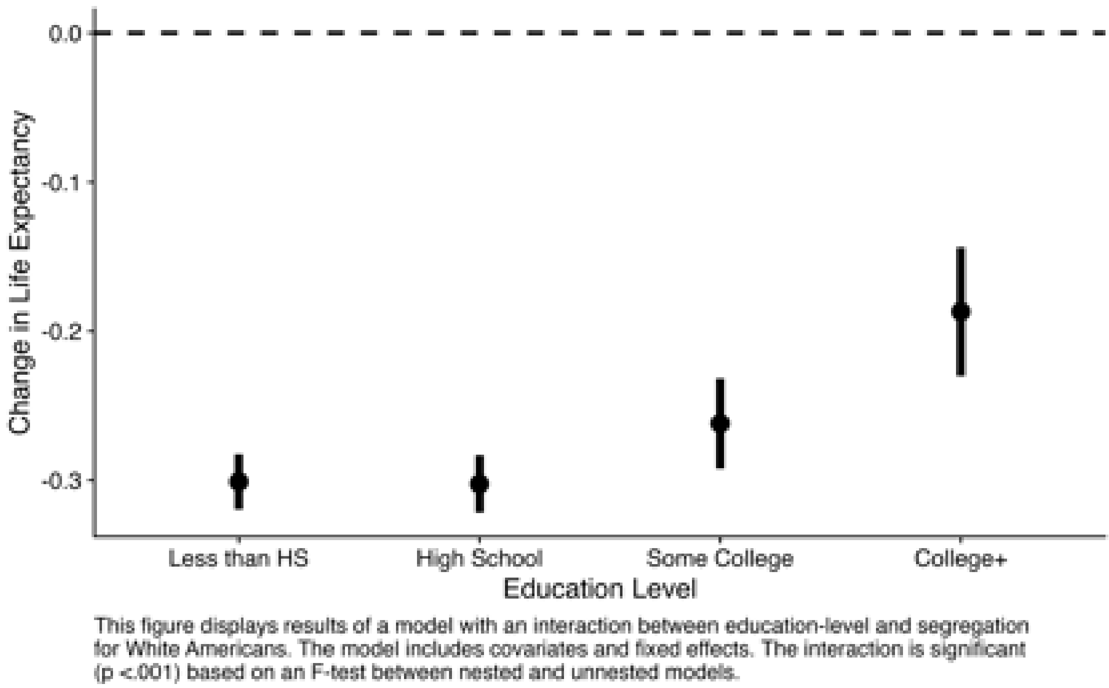


Figure 5: IV Estimates of the Effect of Segregation on Longevity By Education (White)

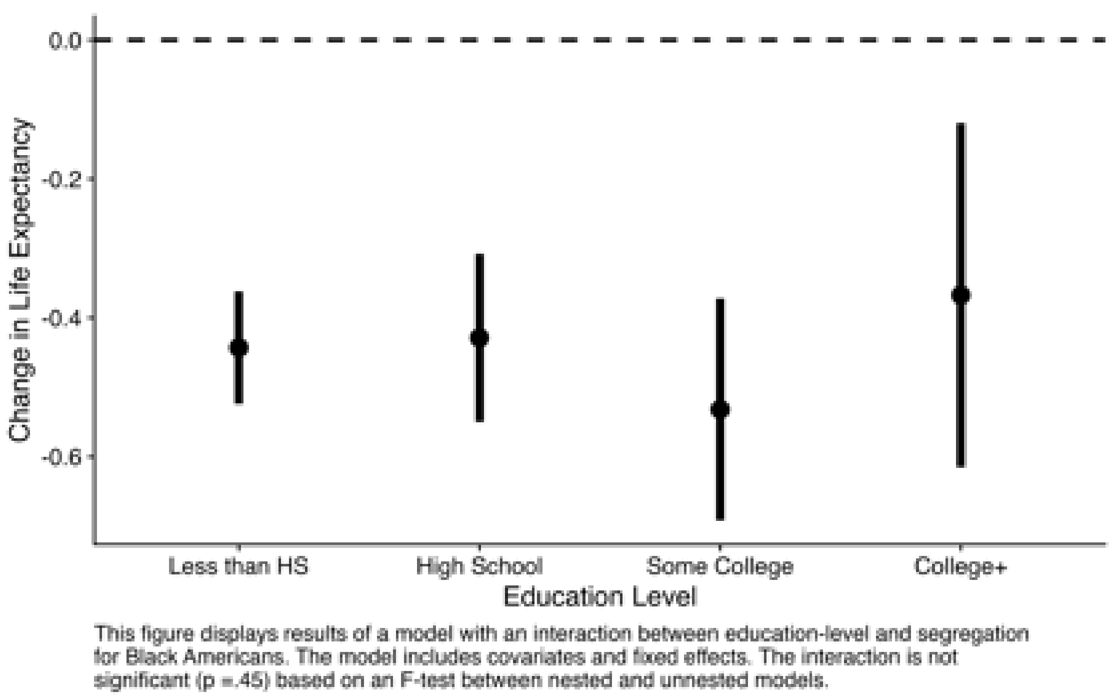


Figure 6: IV Estimates of the Effect of Segregation on Longevity By Education (Black)

Table 1: Representativeness of Numident Analytic Sample V Full Sample

Variable	Mean (All)	Mean (Sample)	diff	p.value
Age at Death	82.03	82.00	0.03	0.00
Male	0.44	0.44	0.00	0.39
Black	0.05	0.05	0.00	0.00
Education (Years)	10.29	10.38	-0.08	0.00
Married (In 1940)	0.49	0.49	0.00	0.00
Migrated (Birth County - Death County)	0.29	0.29	0.00	0.00
Reside in South at Death	0.31	0.29	0.02	0.00
County D at Death	0.53	0.54	-0.01	0.00
County Population	772326.71	849894.69	-77567.98	0.00
County Pr. Black	0.11	0.11	0.00	0.00
N	2415272.000	2094017.000	-	-

This table compares the descriptive statistics of the full Numident sample to the analytic sample subset of the sample used in analysis.

Table 2: Descriptive Statistics of Analytic Sample By Race

	Black	White
Age at Death	81.50	82.03
Male	0.42	0.44
Education (Years)	7.65	10.52
Married (In 1940)	0.54	0.48
Migrated (Birth County - Death County)	0.25	0.29
Reside in South at Death	0.58	0.28
County D at Death	0.60	0.54
County Population	1103625.64	836238.79
County Pr. Black	0.25	0.10
N	106945.00	1987072.00

This table presents descriptive statistics for the analytic sample by racial group.

Table 3: Estimates of the Association Betwen Segregation and Longevity

	Black	Black Controls	White	White Controls
D	-0.084***	-0.084***	-0.072***	-0.072***
	(0.001)	(0.001)	(0.000)	(0.000)
Male		-0.976***		-0.927***
		(0.044)		(0.010)
Education		0.040***		0.055***
		(0.005)		(0.001)
Migrated		-0.456***		0.013+
		(0.037)		(0.008)
Married in 1940		0.107**		-0.048***
		(0.033)		(0.008)
Num.Obs.	106945	106945	1987072	1987072
R2	0.381	0.391	0.350	0.357
Birth Year	X	X	X	X
Birth State	X	X	X	X
Urban-Rural Code	X	X	X	X
Occupation	-	X	-	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$
Heteroskedasiticty Robust Standard Errors in parentheses.

Table 4: Estimates of the Effect of Segregation on Longevity

	Black 1st Stage	Black	Black Controls 1st Stage	Black Controls	White 1st Stage	White	White Controls 1st Stage	White Controls
D		-0.040***		-0.044***		-0.036***		-0.029***
		(0.003)		(0.004)		(0.001)		(0.001)
Male			0.438***	-1.012***			0.391***	-0.945***
			(0.109)	(0.045)			(0.027)	(0.010)
Education			0.023+	0.035***			-0.088***	0.058***
			(0.012)	(0.005)			(0.004)	(0.001)
Migrated			-2.009***	-0.292***			3.372***	-0.138***
			(0.097)	(0.039)			(0.022)	(0.008)
Married in 1940			-0.344***	0.121***			-0.518***	-0.023**
			(0.082)	(0.033)			(0.023)	(0.008)
Died in South			-5.853***	0.012			5.077***	0.195***
			(0.116)	(0.054)			(0.031)	(0.010)
Log(Governments)	4.134***		2.966***		4.544***		4.943***	
	(0.036)		(0.039)		(0.012)		(0.013)	
Revenue Share	78.232***		76.845***		31.002***		38.153***	
	(0.603)		(0.558)		(0.641)		(0.612)	
Num.Obs.	106945	106945	106945	106945	1987072	1987072	1987072	1987072
R2	0.528	0.373	0.547	0.384	0.351	0.342	0.368	0.347
Birth Year	-	X	-	X	-	X	-	X
Birth State	-	X	-	X	-	X	-	X
Urban-Rural Code	-	X	-	X	-	X	-	X
Occupation	-	-	-	X	-	-	-	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

This table describes the first-stage models and IV estimates of the effect of segregation on longevity. Heteroskedasiticty Robust Standard Errors in parentheses.